GHOSTCITE: A Large-Scale Analysis of Citation Validity in the Age of Large Language Models

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Citations provide the basis for trusting scientific claims; when they are invalid or fabricated, this trust collapses. With the advent of Large Language Models (LLMs), this risk has intensified: LLMs are increasingly used for academic writing, yet their tendency to fabricate citations ("ghost citations") poses a systemic threat to citation validity.

To quantify this threat and inform mitigation, we develop CITEVERIFIER, an open-source framework for large-scale citation verification, and conduct the first comprehensive study of citation validity in the LLM era through three experiments built on it. We benchmark 13 state-of-the-art LLMs on citation generation across 40 research domains, finding that all models hallucinate citations at rates from 14.23% to 94.93%, with significant variation across research domains. Moreover, we analyze 2.2 million citations from 56,381 papers published at top-tier AI/ML and Security venues (2020–2025), confirming that 1.07% of papers contain invalid or fabricated citations (604 papers), with an 80.9% increase in 2025 alone. Furthermore, we survey 97 researchers and analyze 94 valid responses after removing 3 conflicting samples, revealing a critical "verification gap": 41.5% of researchers copy-paste BibTeX without checking and 44.4% choose no-action responses when encountering suspicious references; meanwhile, 76.7% of reviewers do not thoroughly check references and 80.0% never suspect fake citations. Our findings reveal an accelerating crisis where unreliable AI tools, combined with inadequate human verification by researchers and insufficient peer review scrutiny, enable fabricated citations to contaminate the scientific record. We propose interventions for researchers, venues, and tool developers to protect citation integrity.

I. INTRODUCTION

Citations serve as a trust mechanism that justifies scientific claims [1], [2]: readers assume that cited work exists and supports the claims being made, and rarely verify that it does [3]. However, when a citation is invalid, this trust mechanism collapses. The advent of AI-assisted academic writing has introduced a novel threat to this scholarly trust infrastructure [4]. This threat is especially acute in security research, where literature reviews underpin threat modeling, vulnerability taxonomy construction, and reproducibility of offensive and defensive techniques; a fabricated citation in a security paper can misdirect defensive investment, introduce phantom prior art, or undermine the evidentiary chain of a claimed attack or mitigation. Researchers increasingly leverage LLMs to synthesize literature reviews, curate related work, or compile reference lists. When prompted for citations, these systems do not look up references. Instead, leveraging their inherent generative nature [5], they *synthesize* references by combining statistically correlated tokens (real author names, plausible-sounding titles, and prestigious venue names) into citations that appear authentic despite being entirely fabricated. This phenomenon commonly described as "ghost citations" [6]–[8], representing a qualitatively new category of academic misconduct: high-fidelity fabrications that exploit the format-compliance of bibliographic conventions to evade detection. Such ghost citations pose a systemic threat to the integrity of scholarly communication. The emergence of ghost citations has caught the attention of prestigious conferences and academic governing bodies. For instance, ICLR 2026 chairs issued guidelines warning against AI-generated content, including fabricated citations [9], and major publishers such as IEEE have established policies requiring disclosure of AI usage in manuscripts [10]. Recent reports also document hallucinated citations in conference submissions and accepted papers, as well as in domain-specific analyses [11]–[13]. Yet despite growing awareness, we lack systematic answers to three fundamental questions: *Q1: How prevalent is this problem? Q2: Has it already contaminated the published scientific record? Q3: And why do verification mechanisms fail?*

Research Gap Prior work and policy discussions highlight the risks of ghost citations [4], [9], [14], but to answer the above questions, three practical gaps remain. First, scalable detection is difficult because citations are heterogeneous [15], and need to be verified against incomplete and inconsistent bibliographic databases; this makes automated, high accuracy verification challenging at scale. Second, we still lack large-scale, empirical measurement of how prevalent ghost citations are in both LLM outputs and the published record. Third, we lack a clear explanation for why verification practices fail: when and why authors or reviewers skip checks, and how those behaviors allow fabricated citations to persist. Fourth, the security community lacks evidence on how ghost citations specifically compromise trust in security literature, despite the field's reliance on precise attribution for tracking threat evolution and validating defensive claims. These gaps motivate

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our comprehensive study.

Our Study. In this paper, we develop CITEVERIFIER, an open-source framework for large-scale citation verification. It combines robust reference parsing with a cascaded, multi-source retrieval strategy (local bibliographic indexes, academic databases, and web search). To achieve scalability and robustness against citation format variations, it leverages caching, LLM-assisted re-parsing, and a similarity-based matching scheme [16] to verify citations at scale (Section III). Using CITEVERIFIER, we conduct the first comprehensive, multi-dimensional analysis of citation validity in the age of LLMs, addressing the three research questions outlined above through complementary studies (Section IV).

LLM Benchmark (Q1). We systematically evaluated 13 state-of-the-art LLMs (including GPT-5, Claude-4, and others) across 40 computer science research domains aligned with arXiv CS subject classes [17], yielding a benchmark dataset of 375,440 LLM-generated citations from 22,800 API interactions at an approximate cost of $800. This benchmark shows that all tested models hallucinate citations, with rates from *14.23%* (DeepSeek) to *94.93%* (Hunyuan). It also reveals strong domain sensitivity (51.39 percentage point spread), and some domains exhibit substantially higher hallucination rates than others (Section V).

Archival Analysis (Q2). We ethically collected [18] 56,381 papers from eight AI/ML and Security venues (NeurIPS, ICML, IJCAI, AAAI, IEEE S&P, USENIX Security, CCS, NDSS) spanning 2020 to 2025, selected by CSRankings [19]. After employing CITEVERIFIER to automatically detect citation validity across the 2,199,409 citations in these papers, we flagged 2,530 citations as potentially invalid. To ensure accuracy, a 16-person research team manually verified the flagged citations over about one month, confirming *739 citations in 604 papers (1.07%)* as definitely invalid with severe metadata errors or untraceable titles. We also observed an *80.9% increase* in invalid citation rates in 2025 over 2020–2024 averages, suggesting a rapidly accelerating problem, and identified *repeated invalid citations* that recur in up to *16* independent papers, indicating error propagation (Section VI).

User Study (Q3). We conducted a survey of active researchers across career stages, recruiting participants via social media and targeted emails to 300 randomly sampled authors and PC members from the above 8 venues; email addresses were manually collected from public sources. After removing 3 conflicting samples, we obtained 94 valid responses. Among respondents who answered the AI-use question (n=86), *87.2%* use AI tools for research tasks and *86.7%* of AI users claim to "always verify" AI-generated citations, yet behavioral data reveals a *verification gap*: 41.5% copy-paste BibTeX without checking, and 44.4% take no action when encountering suspicious fabricated references. Among reviewers (n=30), 76.7% do not thoroughly check references, and 80.0% never suspect fake citations in submissions, indicating a trust-by-default norm that allows ghost citations to bypass both author and reviewer checks. At a broader level, 74.5% of researchers view peer review as ineffective at catching metadata errors, while 70.2% strongly support automated checks (Section VII).

Our findings reveal a concerning trend: *LLMs are unreliable for citation generation*, *untraceable citations have already penetrated the scientific record*, and *human verification mechanisms systematically fail* to catch them. For the security community specifically, ghost citations constitute a new form of computing abuse that erodes the evidentiary foundations of threat research, pentesting methodologies, and vulnerability disclosure practices. Our work follows the tradition of meta-science measurement studies that expose structural problems in research practices [20]–[23]. Just as prior work revealed reproducibility crises and evaluation pitfalls [20], [22], we identify an emerging threat to the integrity of scholarly citation, one that demands immediate attention as AI adoption accelerates. We strongly argue that the scientific community should treat ghost citations as a systemic threat to scholarly integrity. Addressing this challenge requires immediate and coordinated efforts from all stakeholders, including publishers, reviewers, and the researchers themselves.

Contributions. Our contributions are as follows:

1) **Technical baseline for citation verification.** We frame ghost citations as a systemic threat to research integrity, particularly in security, and develop GHOSTCITE, an open-source framework that verifies citations at scale, providing a replicable implementation example for venues, publishers, and future work.
2) **Large-scale benchmark of LLM citation accuracy.** We evaluate *13* LLMs across *40* domains on *375,440* citations, showing hallucination rates of *14.23–94.93%* with significant research domain sensitivity.
3) **Empirical analysis of the published record.** We analyze *2.2M* citations from *56,381* papers (2020–2025), finding *604/56,381 (1.07%)* with invalid citations, an *80.9%* increase in 2025, and evidence of propagation (up to *16* repeated errors).
4) **Behavioral analysis of verification practices.** Through a survey of *94* valid responses (97 total, 3 removed), we identify a *verification gap* across roles: *41.5%* copy-paste BibTeX without checking, *76.7%* of reviewers do not thoroughly check references, and 74.5% view peer review as ineffective at catching metadata errors while 70.2% strongly support automated checks.

II. BACKGROUND AND RELATED WORK

Citations serve as a trust mechanism that justifies scientific claims; the validity of citations is therefore central to the integrity of scholarly communication [24]. When citations are invalid, this trust breaks down: readers may be misled by references that do not support, or do not even exist to support, the claims being made. Such fabrications are commonly termed "ghost citations" [6]–[8]. To understand and address this threat, we first need to clarify three fundamental questions: What exactly are ghost citations, how did they arise, and why do they pose such a risk to scientific research?

A. *Ghost Citations in the Wild*

The problem of erroneous citations predates the era of generative AI. Early citation-analysis studies reported that citation errors, including incorrect author names, publication years, page numbers, and even entirely non-existent sources, have been a persistent problem in academic publishing [25]. Furthermore, Simkin and Roychowdhury [26] estimated that a substantial fraction of citations are copied from secondary sources without consulting the original work, propagating errors through the literature. These human errors, while problematic, typically arose from carelessness or negligence rather than intentional fabrication.

However, the emergence of LLMs fundamentally transformed the nature of citation errors, introducing a qualitatively different phenomenon: the wholesale fabrication of references that never existed [14]. Unlike human mistakes, LLM-generated citations are systematically produced through autoregressive token prediction [5], [27]. When generating academic text, LLMs identify high-probability token clusters associated with a research field (authors like "Vaswani," venues like "NeurIPS," and domain-specific terminology) and "stitch" them together into plausible but non-existent citations. As Bender et al. [28] claimed, LLMs function as "stochastic parrots" that prioritize the structure of language over its truth. Because academic citations follow rigid syntactic templates, LLMs can effortlessly mimic this surface form, producing fabricated references indistinguishable from authentic ones.

The phenomenon of LLM hallucination has been extensively studied [29]–[31], with factuality benchmarks such as TruthfulQA [32] and HaluEval [33] quantifying these failures. As generative AI tools proliferated in academic contexts [34]–[36], anecdotal evidence of fabricated citations accumulated in online forums [6] and academic blogs [7], [8]. Detection efforts have emerged: GPTZero reported hallucinated citations in both ICLR 2026 submissions and NeurIPS 2025 accepted papers [11], [12], while watermarking [37] and text writing-style analysis [38] offer complementary strategies. Nevertheless, these tools primarily target AI-generated prose rather than the factual accuracy of citations themselves. Recent reports and empirical studies further underscore the problem's breadth; ACL-focused analyses identified hundreds of hallucinated references [13], and domain-specific studies in librarianship documented hallucinated citations in ChatGPT outputs [14]. Nevertheless, we still lack systematic, large-scale analysis of ghost citation prevalence across models and the published literature, as well as a clear understanding of why verification behaviors fail to prevent their propagation.

B. *The Risk to Scientific Research*

The spread of ghost citations goes beyond mere typographical or formatting errors; it constitutes a systemic threat to the integrity of scholarly communication, with the capacity to undermine scientific research at scale. We analyze these risks across three key stakeholder groups.

Impact on Researchers. In normal scholarship, citations ground claims in verifiable prior work and help researchers navigate the literature with confidence [1], [2]. Ghost citations subvert this process by inserting fabricated references that appear legitimate, misdirecting research efforts and wasting resources on phantom threads; when they enter literature reviews or used as starting points for new investigations, they distort scientific truth and propagate misinformation through the research pipeline. In this situation, ghost citations create *Epistemic Pollution*: non-existent studies are cited to support claims, creating an illusion of evidential support [24].

Impact on Reviewers and Publishers. The academic ecosystem operates on a foundational trust protocol: we assume authors work in good faith, and reviewers rarely verify the work exhaustively [39]. Plausible-looking LLM citations exploit this trust gap; verifying 30–50 references per paper would impose prohibitive burdens on the already-strained peer review process. Our own manual verification required a 16-person research team working for about one month to review 2,530 flagged citations (Section VI), demonstrating that thorough checking demands substantial human time and coordination. Publishers and conference organizers lack scalable infrastructure to detect fabricated citations, leaving the burden entirely on human reviewers who have neither the time nor the tools to systematically validate references.

Impact on the Research Community. Research impact analysis relies on the citation graph (nodes = publications, edges = citations) [40]. Ghost citations inject phantom nodes and invalid edges, undermining citation-based indicators (e.g., Impact Factor, h-index) and network metrics (co-citation, bibliographic coupling), degrading the reliability of quantitative impact assessment. More fundamentally, as fabricated citations accumulate in the published record, they erode the trust infrastructure that underpins scholarly communication. If future LLMs are trained on papers containing ghost citations, these fabrications may be reproduced amplifying errors across generations [41]. Ultimately, widespread ghost citations may trigger a "Crisis of Legitimacy" that forces the research community to shift from presumptive trust to systematic skepticism, imposing unsustainable verification burdens on scientific communication.

C. *Research Gap*

Ghost citations pose systemic risks across all stakeholders in the research ecosystem, from individual researchers to publishers and the broader scientific community. Recent reports and targeted analyses have documented hallucinated citations in conference submissions and accepted papers [11], [12], as well as in ACL-focused studies [13]. Yet despite this growing evidence, three critical research gaps remain unaddressed.

RG1: How can ghost citations be detected at scale? Unlike AI-generated prose detection, citation verification requires cross-referencing against external bibliographic databases, a challenge complicated by the vast heterogeneity of citation formats [15], the incompleteness of existing scholarly indices, and the absence of standardized verification protocols. Major publishers including Nature, Science, and IEEE have issued policies requiring disclosure of AI usage [4], [10], [42], and

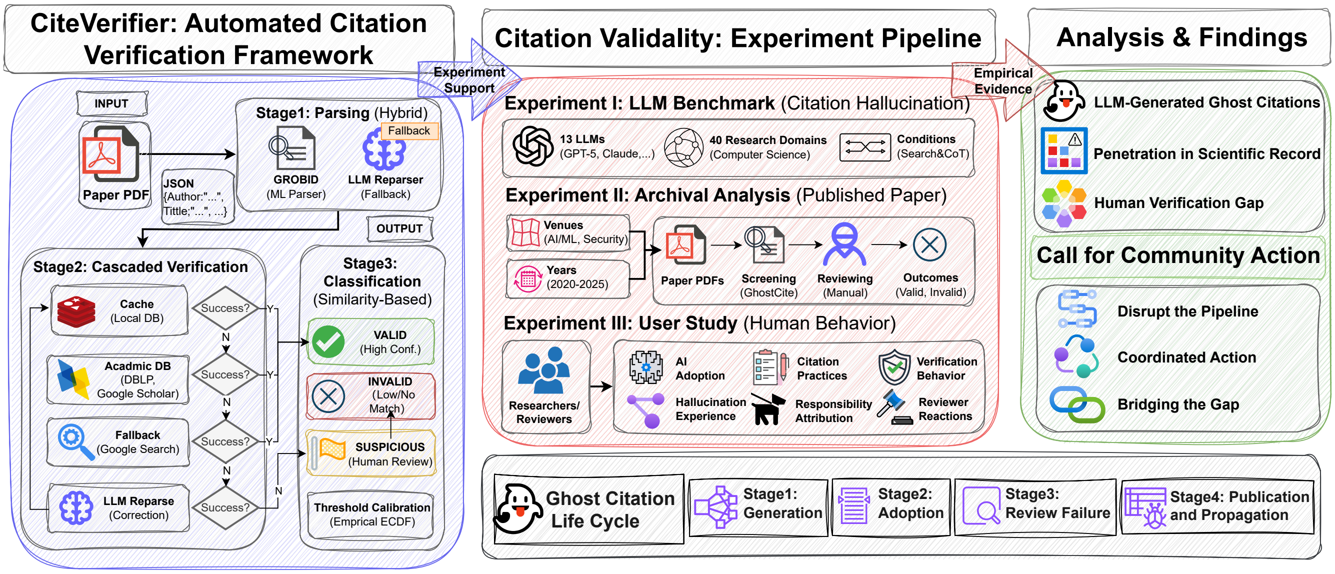


Fig. 1: The framework of CITEVERIFIER and experimental pipeline.

academic conferences such as ICLR 2026 have established explicit guidelines [9]. However, policy frameworks remain fragmented, and no scalable, automated verification infrastructure exists to systematically detect fabricated references across the published literature.

RG2: What is the current prevalence of ghost citations in the published literature? Prior meta-science measurement studies have demonstrated the value of large-scale empirical analysis in exposing systemic issues, from reproducibility crises [20], [21] to evaluation pitfalls in machine learning [43], [44] and citation manipulation practices [45]. Yet no comparable study has systematically quantified the prevalence and characteristics of ghost citations. Existing evidence remains largely anecdotal, lacking the rigorous methodology that is necessary to determine whether ghost citations represent just an isolated incident or a widespread undermining of the whole scientific research.

RG3: Why do verification practices fail to prevent ghost citations from reaching publication? Surveys and guidelines routinely advise researchers to verify AI-generated contents [35], [36], and many researchers report that they do so [46]. Yet the gap between self-reported verification and actual behavior has not been systematically studied: it remains unclear how often researchers rely on AI-generated citations, under what conditions verification is skipped, and which behavioral or cognitive factors allow fabricated citations to persist in the pipeline from model output to published paper.

Our work addresses all three research gaps through conducting the first comprehensive, multi-dimensional analysis of ghost citations in the LLM era, combining technical development with empirical measurement and behavioral analysis. We develop CITEVERIFIER, an automated citation verification pipeline with robust parsing, cascaded multi-source retrieval, and calibrated similarity matching to achieve the high accuracy and support high-throughput validation. Leveraging CITEVERIFIER, we conduct three complementary studies: (1) a systematic benchmark of citation hallucination rates across 13 LLMs under varying conditions (RG1; Section V); (2) a large-scale audit of 56,000+ published papers to quantify real-world prevalence (RG2; Section VI); and (3) a user study investigating the behavioral factors that enable fabricated citations to reach publication (RG3; Section VII).

III. AUTOMATED VERIFICATION FRAMEWORK

To systematically analyze the prevalence and impact of ghost citations, we first need a reliable method to identify them at scale. To this end, we developed CITEVERIFIER, an automated citation verification framework designed to robustly identify invalid citations across large corpora of academic papers. In this section, we describe the design and implementation of our CITEVERIFIER framework, detailing how it addresses the significant challenges of citation verification to enable our subsequent analyses.

A. Framework Design

Verifying citations at scale raises several technical and practical challenges; we summarize these challenges below, then state the design goals we set to address them.

Design Challenges.

- *Parsing heterogeneity*: academic citations exhibit enormous format diversity, varying across venues, disciplines, and even individual papers; PDF extraction introduces additional noise through OCR errors, non-standard fonts, so a robust pipeline needs to tolerate parsing imperfections without generating excessive false positives.
- *Database coverage*: no single bibliographic database provides comprehensive coverage; different sources have different profiles, so absent results are often ambiguous, motivating a multi-source strategy.

- *Verification scalability*: verifying millions of citations requires a well-engineered framework design, such as rate limiting, cache reuse, and concurrent requests, to achieve practical throughput without excessive costs.

Design Goals. To address these challenges, we set the following design goals: (1) This system should provide parsing robust to format heterogeneity that recovers from extraction failures. (2) It should provide verification leveraging multiple sources in a cost-effective order and scaling to large corpora. (3) It should remain robust to diverse citation formats to prevent legitimate references from being over-flagged as invalid With these goals in mind, we developed CITEVERIFIER, an automated citation verification framework.

Design Overview. With these challenges and goals, we designed CITEVERIFIER as a modular, cascaded verification pipeline, as illustrated in Figure 1. It follows a retrieval-first, error-tolerant pipeline to verify citations at scale. It parses noisy reference strings into structured metadata, queries a cascaded set of sources, and then classifies validity using similarity-based matching to tolerate minor metadata variations. The cascade prioritizes low-cost, high-precision sources and falls back to broader search only when needed, while an LLM reparser serves as a last-resort correction for malformed inputs. This design balances coverage and precision, ensuring that the system catches fabricated references without over-flagging legitimate citations.

B. Framework Implementation

CITEVERIFIER is implemented as a three-stage pipeline: *parsing*, *verification*, and *classification*; we describe each stage below.

Stage 1: Reference Parsing. The parsing stage aims to extract structured metadata from raw citation strings while robustly handling OCR errors and diverse citation formats. We employ a hybrid parser that combines standard PDF extraction with an LLM-based fallback: when the initial parsing leads to a verification failure, we re-parse and then re-run the verification chain with LLM-extracted metadata, ensuring high accuracy across varied parsing scenarios.

Stage 2: Cascaded Verification. With structured metadata extracted, the verification stage checks citation validity using a cascaded strategy with four verification methods executed sequentially:

1) *CacheVerificationStrategy.* We first query a local SQLite database of previously verified citations. This cache stores results from prior searches, enabling efficient re-verification and reducing API costs. The cache is keyed by normalized title and returns stored verification results along with matched external reference metadata.
2) *AcademicDBVerificationStrategy.* For uncached entries, we query multiple bibliographic databases in cascade. We prioritize local academic databases for efficiency and reliability, then extend to broader search platforms when needed. The query is constructed from the citation title, and returned results are parsed to extract title, authors, year, and venue metadata. We store all results in the cache for future queries.
3) *WebSearchFallbackStrategy.* If the academic databases fail to find the reference, we fall back to general web search. This broader search covers recent papers not yet indexed by academic databases, technical reports, preprints, and non-traditional academic content.
4) *LLMReparseStrategy.* If all prior strategies fail to find a match, we hypothesize that parsing errors may have corrupted the search query. The system invokes the LLM reparser to re-extract metadata from the raw string, then re-executes the verification chain with the corrected metadata. This addresses cases where minor parsing errors, ensuring high accuracy even with noisy inputs.

Each strategy returns either a `VerificationResult` (if verification succeeds or definitively fails), or triggers the next strategy in the cascade.

Stage 3: Similarity-Based Classification. After retrieving candidate matches from the verification stage, we classify the citation as `Valid` or `Invalid` based on the similarity between the input citation and retrieved candidates. In this part, we only focus on title similarity, as our preliminary goal is to identify citations that are untraceable or even non-existent (i.e. ghost citations), rather than citation errors as reported in prior work [3], [26], The classfication is based on the highest similarity score among all retrieved candidates, using a empirical calibrated threshold θ to determine validity: if the maximum similarity exceeds θ, we classify the citation as `Valid`; otherwise, it is classified as `Invalid`.

C. Implementation Details

We implement CITEVERIFIER in Python with asynchronous I/O (asyncio) for concurrent API requests, enabling efficient parallel processing of multiple citations. Below, we detail each key component.

Parsing and Verification Services. For parsing, we use GROBID [47] as the primary parser and a Qwen3-Flash [48] based LLM reparser for cases with parsing failures, extracting structured metadata via a JSON schema. For verification, we employ multiple sources: we construct a local DBLP database by parsing the DBLP XML dump, enabling efficient local verification queries; we also access Google Scholar and Google Search through the ScrapingDog API proxy [49] for searching recent or non-indexed papers.

Similarity Computation and Threshold Selection. For similarity-based classification, we normalize titles and compute Levenshtein distance [16] (using the highest similarity among all retrieved candidates). We set the classification threshold to $\theta = 0.9$, roughly corresponding to a one–two word difference in a long title. We validate this choice based on the subsequent experiments in Section V and Section VI, using the empirical CDF of title similarity for real-paper citations and LLM-generated citations (Figure 2). Real-paper similarities remain near 1.0: only 0.4% fall at or below 0.9, and the small tail below 1.0 is largely attributable to noise

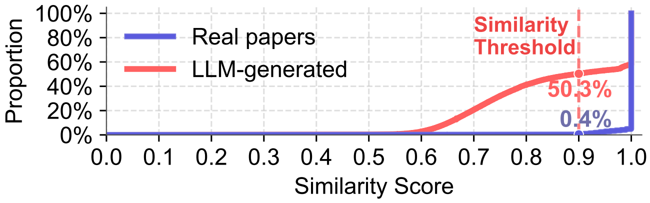


Fig. 2: ECDF of title similarity scores: LLM-generated vs. real-paper citations.

(OCR errors, hyphenation, minor typos). In contrast, LLM-generated citations rise gradually, with 50.3% at or below 0.9, validating $\theta = 0.9$ as a threshold that preserves high recall for real papers while filtering low-similarity hallucinations.

Practical Considerations. Key implementation choices include semaphore-based rate limiting (default: 10 concurrent requests) to respect API quotas; batch processing of entire directories with progress tracking and incremental result export; and CSV export with columns for final status, diagnosis, similarity scores, and matched external reference metadata. Taken together, these design and engineering choices make CITEVERIFIER a practical, scalable, and robust verification system for large-scale citation analysis, enabling both our subsequent experiments and real-world deployment.

IV. EXPERIMENT DESIGN

To measure the prevalence of ghost citations and understand their origins across the research ecosystem, we conduct three complementary experiments: an LLM Benchmark to quantify ghost-citation generation by state-of-the-art LLMs; an Archival Analysis to measure real-world prevalence; and a User Study to understand the human factors that allow ghost citations to persist. Below, we describe each in detail.

A. Experiment I: LLM Benchmark

In this experiment, we systematically evaluate 13 state-of-the-art LLMs spanning major vendors and diverse architectural approaches (see Table V for the complete list), assessing their ability to generate valid academic citations.

Setup. All models were accessed via a third-party API aggregation platform (OpenRouter) [50] for consistent interfaces. We constructed a taxonomy of 40 computer science research domains based on arXiv CS subject classes [17] (full list in the Section B). For each model-domain combination we use a standardized prompt to generate citations within that domain; the prompt asks the model to produce a fixed number of domain-specific references in a strict JSON schema (author, title, venue, year, type), enabling consistent parsing and verification (the full prompt is provided in Section C). **Conditions and verification.** We varied two factors: batch size (10, 20, or 30 citations per prompt) and whether to enable online search with chain-of-thought prompting. For each model-domain-batch-search combination, we generate 120 citations (30×4 interactions, 20×6 interactions, or 10×12 interactions). In total, we obtained 375,440 citations from 22,800 API interactions, with total API costs amounting to ≈ 800 USD. All generated citations were verified with CITEVERIFIER and classified as VALID or INVALID. Unparseable outputs were flagged as format errors and only included in format compliance statistics, not hallucination rates.

Limitations of experimental setup. We acknowledge that our benchmark design—prompting models to generate fixed-size blocks of domain-specific citations—does not fully mirror real-world researcher workflows, where citations are typically produced in the context of drafting specific arguments rather than via bulk generation requests. This design choice prioritizes internal validity (ensuring fair cross-model comparison under identical conditions) over external validity (generalizability to naturalistic use). Accordingly, our hallucination rates should be interpreted as a controlled baseline rather than an estimate of real-world prevalence, and we discuss this limitation further in Section VIII.

Auxiliary Experiment: Citation Validity Judgment. To assess whether LLMs can reliably evaluate citation validity, we conducted an auxiliary experiment where each model was prompted to verify 100 bibliographic entries with known ground truth (50 valid, 50 invalid) from our archival analysis (Section VI), one entry per prompt, and classify each as VALID or INVALID. The prompt is provided in Appendix D.

B. Experiment II: Archival Analysis

To quantify the real-world penetration of ghost citation into the scientific record, we analyze published papers from top-tier venues. We target two research communities: *Security* (IEEE S&P, USENIX Security, ACM CCS, NDSS) and *AI/ML* (NeurIPS, ICML, IJCAI, AAAI), focusing on venues that are central to LLM research and have rigorous peer review [19]. Data span 2020–2025 (pre-LLM 2020–2022, post-LLM 2023–2025) [51] for longitudinal analysis. We collected all accepted papers (main, workshop, short, poster) in PDF from official proceedings using ethically compliant access and stewardship practices (see Section IX), yielding a total of 56,381 papers. To ensure the accuracy of our analysis and to avoid false positives (where valid citations are misclassified as invalid), we employ a rigorous three-stage process:

- *Step 1: Automated Examination.* All 56,381 papers were processed through CITEVERIFIER, extracting citations, and flagging potentially problematic citations as invalid, meaning their title falls below the similarity threshold $\theta = 0.9$.
- *Step 2: Manual Verification.* Sixteen trained research assistants manually reviewed all flagged citations over approximately one month. Each citation was independently checked at least twice before being classified into one of three categories: *Non-Academic Source* (websites, blogs, repositories), *Valid* (verified through extensive manual search), or *Invalid* (incorrect metadata or untraceable).
- *Step 3: False Negative Estimation.* We sampled 400 citations from the "valid" pool for manual review. This sample size was determined using standard sample-size estimation for proportions: with an expected false-

negative rate near zero, a 95% confidence level, and a 5% margin of error, approximately 384 samples are required; we rounded up to 400 to ensure robustness ocite-cochran1977sampling. No additional invalid citations were found, suggesting high recall.

Two expert researchers subsequently reviewed all manually classified invalid citations to ensure accuracy, and further subdivided them into two subcategories: *error citation* and *ghost citation* (untraceable citations cannot be found even with extensive manual search). Some untraceable citations appear clearly fabricated (e.g., titles stitched together from high-probability phrases); others simply do not exist in standard bibliographic indexes, and both align with the definition of "ghost citations" in our study.

C. Experiment III: User Study

To understand the human factors enabling ghost citations to reach publication, we conducted a survey of active researchers across career stages and research areas, recruiting participants via social media and targeted emails to 300 randomly sampled authors and PC members from the above 8 venues; email addresses were manually collected from public sources. The survey covered four dimensions: AI adoption, citation practices, author-level verification behavior, and reviewer-level verification behavior, and included paired reverse-worded questions to flag inconsistent responses as a data-quality check (the full survey is provided in Section J). Following IRB approval, the online survey provided full study details and opt-out options; it took 10–15 minutes, was voluntary and anonymized, and obtained informed consent without collecting personally identifiable information. Responses were stored and analyzed in de-identified form with access control.

Data analysis. Quantitative responses were analyzed using descriptive statistics (frequencies and percentages). For the open-ended comment question (Q59), two researchers independently performed thematic coding to identify recurring themes; disagreements were resolved through discussion. To ensure response consistency, we included paired reverse-worded items (Q38 and Q39): respondents who simultaneously endorsed risky behavior ("often copy-paste without checking") and diligent behavior ("meticulously verify every field") were flagged as inconsistent and excluded (3/97), yielding $N = 94$ valid responses for analysis.

V. Measuring the Unreliability: LLM Benchmark

We benchmark 13 state-of-the-art LLMs on citation generation tasks, leveraging CiteVerifier to verify each output against authoritative bibliographic sources. Our goal is to establish the prevalence of citation hallucination at its source and identify factors that may influence reliability.

Statistics Overview. Despite the structured JSON format specified in our prompts, a notable fraction of models struggled to adhere to the requested format, resulting in unparseable outputs. In total, we obtained 20,653 well-formed JSON outputs from 22,800 interactions (90.58% format compliance) and successfully extracted 331,809 citations from the expected 375,440 (88.38% extraction rate). Of these extracted citations, 164,933 (49.71%) were verified as valid, while 166,876 (50.29%) were invalid. We treat invalid citations as fabricated and manually checked two random samples (400 valid, 400 invalid). Valid accuracy was 100%, and invalid accuracy was 98% (392/400); the remaining 8 cases were non-paper sources (4), out-of-domain works (1), or poorly indexed items (3), suggesting relatively low false positive rates and confirm the high precision of the verification pipeline.

TABLE I: LLM performance on citation generation, ranked by hallucination rate (lower is better).

Model	JSON (%)	Citations	Halluc. (%)	95% CI
DeepSeek	97.67	27,973	14.23	±1.65
GLM-4.5	97.22	27,835	21.25	±1.94
Claude 4	100.00	28,800	21.84	±1.93
Qwen-3	99.38	28,546	23.52	±1.99
Kimi	89.09	24,955	41.86	±2.44
Llama 4	97.33	27,925	45.84	±2.36
GPT-5	97.90	28,366	50.92	±2.36
Gemini	95.80	27,102	59.47	±2.34
ERNIE	95.06	27,332	71.90	±2.15
Seed	48.75	11,090	78.64	±2.74
Grok 4	99.03	28,627	79.98	±1.88
Phi-4	93.35	27,164	87.47	±1.60
Hunyuan	62.90	16,094	94.93	±1.29

Halluc.: Hallucination rate, the percentage of citations verified as invalid. **95% CI:** Confidence interval margin (±) for the hallucination rate.

A. Overall Hallucination Rates

Table I presents both JSON format compliance and citation hallucination rates for each model. LLMs exhibit significant variance in citation validity, with hallucination rates spanning from 14.23% (DeepSeek) to 94.93% (Hunyuan), a roughly 6.7× difference. This wide performance gap reveals that citation generation capability is far from uniformly developed across LLM vendors, and users cannot assume that all "state-of-the-art" models are equally reliable for bibliographic tasks.

We further examined factors that might influence hallucination rates, such as online search, chain-of-thought prompting, and batch size. In our experiment, these factors did not show a consistent effect on hallucination rates across models, details of which are provided in Appendix E0a, suggesting that hallucination is a fundamental limitation of the models' bibliographic knowledge rather than an artifact of specific prompting strategies or output quantity.

B. Domain Sensitivity

We also analyzed hallucination rates across 40 research domains, revealing significant *domain sensitivity* in citation generation performance, as shown in Figure 3

Model-Centric View. Within the same model, hallucination rates vary widely across domains. Even the best-performing model, DeepSeek (lowest overall hallucination rate), exhibits marked performance gaps across domains: it achieves strong performance in some domains (e.g., 2.6% in CV) but hallucination rates rise substantially in others (e.g., 52.5% in OH).

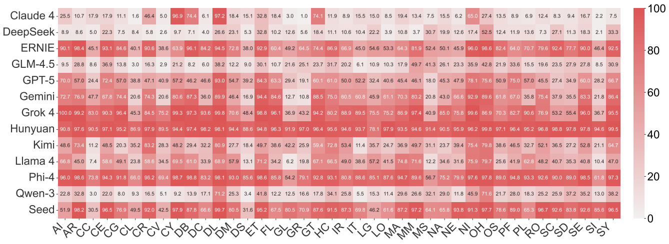


Fig. 3: Model-domain hallucination rate heatmap. x-axis: models, y-axis: domains. Color represents the hallucination rate, the darker the color, the higher the hallucination rate.

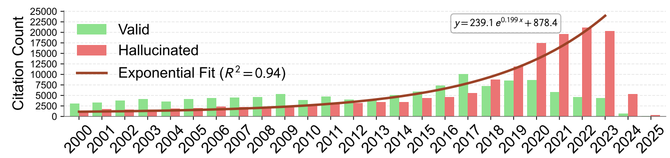


Fig. 4: Temporal distribution of generated citations by publication year (2000–2025); hallucination rates rise steadily and peak at 98.75% in 2025.

No model is uniformly reliable; users cannot assume a low overall hallucination rate implies reliability across all areas.

Domain-Centric View. Within the same domain, models exhibit wide variance in hallucination rates. In Hardware Architecture (AR), Grok 4 approaches near-total failure (99.2%) while DeepSeek remains comparatively accurate (8.6%); in Digital Libraries (DL), frontier models GPT-5 (93.0%) and Claude 4 (97.2%) both show high rates. Specifically, Grok 4 shows a 100% hallucination rate in Artificial Intelligence (AI); manual checks indicate that most outputs are stitched together from the prompted example papers in this domain (prompt details in Appendix C), suggesting overfitting to the prompt rather than generating valid citations. Average hallucination rates across models range from 28.80% (Computation and Language) to 80.19% (Digital Libraries), a 51.39 percentage point gap (Table VII, Appendix F). This variation indicates that researchers in certain subfields face substantially higher exposure to hallucinated citations than other areas.

C. Patterns of Generated Citations

Beyond aggregate hallucination rates, we examine the *characteristics* of generated citations to understand how LLMs fabricate bibliographic information.

Temporal Distribution Analysis. Figure 4 reveals a clear pattern in the publication years of generated citations. Valid citations show a relatively uniform distribution from 2000 to 2014, with a steady increase from 2015 to 2020, followed by a decline (reflecting the models' training data cutoff). In contrast, hallucinated citations exhibit a markedly different pattern: hallucination rates increase steadily with publication year, from 27.61% in 2000 to 98.75% in 2025. We even find an exponential function fit with $R^2 = 0.94$ that models the relationship between publication year and hallucination count. This finding suggests that *LLMs preferentially hallucinate citations with recent publication years*.

Stability, Overlap, and Recurrence. We further analyze the stability of generated citations of each model and the generated citation overlap across models. Valid citations are notably more stable across runs (mean up to 0.58 for DeepSeek and 0.57 for Qwen-3), while hallucinated ones are less stable (mean up to 0.23). Consistent with this, the citations that recur most often are well-known, highly represented papers in model training data, such as "NeRF" in Graphics, "RAG" in NLP, and "U-Net" in Computer Vision; less prominent long-tail works are more prone to hallucination. The detailed stability and overlap analysis is provided in Appendix G

TABLE II: Citation validity judgment accuracy: LLMs evaluated on manually verified ground truth from archival analysis (50 valid + 50 invalid entries). Sorted by overall accuracy.

Model	Valid Recall	Invalid Precision	Accuracy
ERNIE	6/50 (12%)	50/50 (100%)	56/100 (56%)
Hunyuan	19/50 (38%)	29/50 (58%)	48/100 (48%)
Seed	31/50 (62%)	16/50 (32%)	47/100 (47%)
Phi-4	12/50 (24%)	29/50 (58%)	41/100 (41%)
Kimi-K2	14/50 (28%)	26/50 (52%)	40/100 (40%)
GPT-5	8/50 (16%)	28/50 (56%)	36/100 (36%)
Llama-4	19/50 (38%)	16/50 (32%)	35/100 (35%)
DeepSeek	16/50 (32%)	19/50 (38%)	35/100 (35%)
Qwen-3	13/50 (26%)	21/50 (42%)	34/100 (34%)
Gemini	20/50 (40%)	12/50 (24%)	32/100 (32%)
Grok-4	14/50 (28%)	18/50 (36%)	32/100 (32%)
GLM-4.5	19/50 (38%)	14/50 (28%)	33/100 (33%)
Claude-4	12/50 (24%)	15/50 (30%)	27/100 (27%)
Average	15/50 (30%)	23/50 (46%)	38/100 (38%)

≥50% better than random guessing;
<50% worse than guessing.

D. LLM Citation Validation Performance

To assess whether LLMs can reliably evaluate citation validity, we conducted an auxiliary experiment where each of the 13 models was prompted to verify 100 bibliographic entries with known ground truth from our archival analysis in Section VI (50 valid, 50 invalid, full prompt is listed in Appendix D). Each model was prompted to classify each entry as VALID or INVALID, with both online search and chain-of-thought prompting enabled in API settings, and we measured their accuracy.

Table II shows that LLMs perform poorly at citation validation, achieving only 38% average accuracy, even lower than random guessing (50%). Only ERNIE exceeded 50% accuracy (56%), but achieved this by aggressively flagging citations as invalid: it correctly identified all 50 invalid citations (100% precision) but misidentified 88% of valid citations as fabricated. This finding is concerning: not only do LLMs hallucinate citations prolifically, they also *cannot reliably verify citations* when prompted to do so. Users cannot solely rely on LLMs to self-correct or validate bibliographic outputs.

LLM Benchmark

- All 13 state-of-the-art LLMs hallucinate citations (*14.23–94.93%*) with significant domain sensitivity.
- LLMs are poor citation validators (*38%* accuracy), even lower than random guessing.

VI. Hunting for Ghosts: Archival Analysis

In this section, We analyze 2.2 million citations from 56,381 papers published at eight top-tier AI/ML and Security venues (2020–2025), aiming to quantify whether ghost citations have penetrated the published scientific record.

Statistics Overview. Table III provides an overview of the 56,381 papers collected from eight top-tier venues across AI/ML and Security domains from 2020 to 2025. These papers covering NeurIPS (20,387), AAAI (13,821), ICML (11,192), and IJCAI (5,535) from AI venues, as well as USENIX (1,915), CCS (1,756), S&P (1,073), and NDSS (702) from Security venues. Applying CiteVerifier to all 56,381 papers, we extracted 2,199,409 citations. After filtering OCR extraction errors, we flagged 2,530 as potentially problematic (matched title similarity below $\theta = 0.9$).

A. Detection of Invalid Citations

To ensure accuracy, we manually reviewed all 2,530 flagged citations, classifying each by: (1) attempting to locate it in bibliographic databases, (2) if found, verifying its metadata, checking if it was low-quality (e.g., typos, metadata errors), and (3) if not found, determining it as untraceable citation. This process took approximately one month, with each citation independently reviewed for consistency. Two experts researcher subsequently reviewed all invalid citations to ensure accuracy. After manual verification, we classified the 2,530 flagged citations into three categories:

- *Non-Academic Source*: citations pointing to non-academic sources (e.g., websites, blogs, repositories);
- *Valid (Manual Match)*: citations verified as valid through extensive manual search;
- *Invalid*: citations confirmed as invalid, including *metadata error* (incorrect metadata such as wrong title, authors, venue) and *ghost citation* (untraceable citations that cannot be found in any major academic database).

Of the 2,530 flagged citations, 490 (19.4%) were non-academic sources, 1,301 (51.4%) were verified as valid through extensive manual search, and 739 (29.2%) were confirmed as invalid (136 error citations and 603 ghost citations). *In total, 604 papers (1.07% of 56,381)* contained at least one invalid citation, with 133 papers (0.24%) containing error citations and 486 papers (0.86%) containing ghost citations, with 15 papers having both types. We present representative examples below to illustrate the rationale behind our classification and the characteristics of each invalid citation type:

Case Study 1: Metadata Error Citation.

[Kenny and Keane, 2019] Eoin M. Kenny and Mark T. Keane. Twin-systems for explaining ANNs using CBR. In IJCAI-I-19, pages 2708-2715, 2019.

This citation appeared in an IJCAI 2021 paper. Searches in Google Scholar and DBLP yielded no results. However, extensive manual searching located the actual paper: "Twin-Systems to Explain Artificial Neural Networks using Case-Based Reasoning: Comparative Tests of Feature-Weighting Methods in ANN-CBR Twins for XAI", with matching authors and venue but a significantly different title. While our system was designed to detect "ghost citations" that are entirely untraceable even fabricated, it also effectively identifies citations with mismatch title, which can mislead readers by pointing to incorrect or unrelated works. Such errors, though not fabricated, still compromise citation integrity and scholarly communication.

Case Study 2: Untraceable Ghost Citation

[3] Heimdallr: Scalable enclaves for modular applications. In 15th USENIX Symposium on Operating Systems Design and Implementation (OSDI 21). USENIX Association, July 2021.

This citation appeared in a USENIX Security 2022 paper with no author information. Extensive manual searches in Google Scholar and DBLP yielded no results. Our researchers further examined the complete OSDI 2021 proceedings and confirmed that no such paper exists.

Case Study 3: Ghost Citation with Clear Fabrication.

[4] Alex Brown and James Wilson. Adaptive watermarking for source code protection. Journal of Software Engineering, 2023.

This citation appeared in a NeurIPS 2025 paper. Searches in Google Scholar, DBLP, and major bibliographic databases yielded no results. The citation format and structure closely resemble typical LLM-generated references, with generic author names ("Alex Brown", "James Wilson") and a plausible-sounding journal title ("Journal of Software Engineering"), showing clear signs of LLM fabrication.

Case Study 4: Ghost Citation Only in Records.

[9] Zhao et al. Detail enhancement analysis based hardware Trojan detection using thermal maps. Computer Applications In Engineering Education, 54(16):862, 2018.

This citation appeared in a USENIX Security 2024 paper. It appears only in Google Scholar citation records, showing a consistent group of researchers citing this work. However, extensive searches yielded no actual paper, with no associated PDF or bibliographic entry in any major academic database (e.g., IEEE Xplore, ACM Digital Library, SpringerLink). The

TABLE III: Papers collected by conference and year (2020–2025).

	Conference	2020	2021	2022	2023	2024	2025	Total
AI/ML	NeurIPS	1,898	2,334	2,834	3,540	4,494	5,287	20,387
	AAAI	1,864	1,961	1,624	2,021	2,865	3,486	13,821
	ICML	1,084	1,180	1,233	1,828	2,610	3,257	11,192
	IJCAI	778	721	862	846	1,048	1,280	5,535
Security	USENIX	156	246	255	416	404	438	1,915
	CCS	147	224	286	290	414	395	1,756
	S&P	103	110	148	198	261	253	1,073
	NDSS	88	86	83	94	140	211	702
Total		**6,118**	**6,862**	**7,325**	**9,233**	**12,236**	**14,607**	**56,381**

volume and page numbers do not align with the journal's actual publication records for that year, and no such paper exists in the journal's archives. We hypothesize that such citations may result from citing internal or restricted-access papers that cannot be publicly accessed or validated. *Nevertheless, they align with our definition of "ghost citations"*.

Case Study 5: Repeated Invalid Citations.

[Hendrycks et al., 2020] Augmix: A simple method to improve robustness and uncertainty under data shift. In ICLR, 2020.

This citation appeared in 16 different papers published at AAAI, IJCAI, and NeurIPS, and has been cited 220 times according to Google Scholar. Yet the actual paper is titled "AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty", published in ICLR 2020. After extensive root cause tracing, we found that OpenReview's cite button returns this incorrect title, which may be the source of this repeated error. *Such viral invalid citations consistently appeared during our manual verification.* We even traced one clear propagation path where a top-tier paper cited a work with erroneous metadata, and subsequently papers copied this incorrect citation, continuing to propagate the mistake across the literature.

B. Overall Distribution of Invalid Citations

We further analyze their distribution across venues and within individual papers, aiming to understand the prevalence and patterns of invalid citations in published research. Our analysis shows that invalid citations are widespread across all research communities, with patterns suggesting both isolated human errors and systematic AI-assisted generation.

Distribution by Venue. Table IV presents the distribution of invalid citations across the eight venues studied, broken down into *Error* (incorrect metadata) and *Ghost* (untraceable) categories. Invalid citations are present across *all venues*, with NeurIPS exhibiting the highest absolute count (391 papers) and NDSS showing the highest proportion (2.56% of papers) with invalid citations. While AI/ML conferences have a much higher absolute number of papers with invalid citations (largely due to their much greater publication volumes), the overall proportion of papers with invalid citations is remarkably similar: 1.08% for AI venues vs. 1.01% for Security venues. This prevalence suggests that citation integrity issues are not confined to any particular research community.

TABLE IV: Invalid citations by conference. (sorted by invalid citation number)

Conf.	Error	Ghost	Invalid	Papers	Rate
NeurIPS	59	332	391	308	1.51%
ICML	22	103	125	104	0.93%
AAAI	21	85	106	86	0.62%
IJCAI	11	42	53	51	0.96%
NDSS	4	19	23	18	2.56%
CCS	11	9	20	20	1.14%
USENIX	6	6	12	11	0.57%
S&P	1	7	8	6	0.56%
Total	**135**	**603**	**738**	**604**	**1.07%**

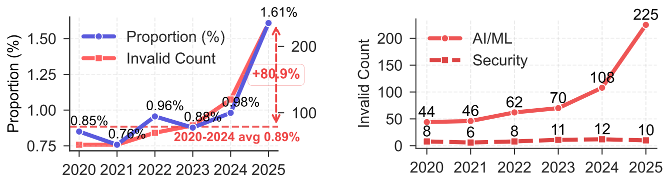
Conf.: Conference acronym. **Invalid:** Error + Ghost citations; **Rate:** percentage of papers with invalid citations.

Distribution per Paper. We additionally analyze the distribution of invalid citations within individual papers. The distribution is heavily right-skewed: 544 papers (88.7%) contain only a single invalid citation, while 68 papers (11.3%) contain two or more invalid citations, with the maximum being *9* invalid citations in a single paper. Our analysis shows that clear-cut AI-induced hallucinated citations often appear in *batches*: papers containing multiple fabricated or entirely non-existent references at once. Thus, a paper exhibiting clusters of invalid citations is a stronger signal of likely AI tool usage, capable of generating groups of spurious "ghost citations" in one go. The presence of multiple invalid citations within a paper is more indicative of systematic, AI-assisted citation generation.

C. Temporal Trends and Propagation

To understand how invalid citations have evolved over time and whether they propagate across papers, we analyze temporal trends from 2020 to 2025 and investigate "repeated invalid citations". Our analysis reveals that invalid citation rates show a *sharp increase in 2025*, This surge aligns temporally with broader shifts in AI-assisted writing practices, but our data alone does not establish causality.

Temporal Trends. Figure 5 visualizes the temporal trends in papers with invalid citations from 2020 to 2025, both overall



(a) Overall Time Trend.

(b) Time Trend by Venue.

Fig. 5: Time trends of papers with invalid citations.

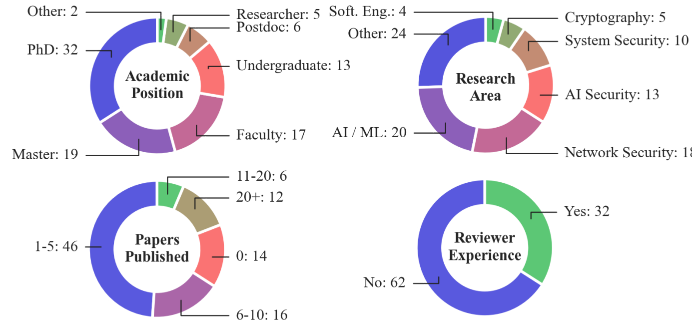


Fig. 6: Survey participant demographics. Soft. Eng.: Software Engineering;

and across individual venues. The proportion of papers with invalid citations remained relatively stable from 2020 to 2024 (ranging from 0.76% to 0.98%), followed by a sharp increase in 2025 to 1.61%. Comparing 2025 to the 2020–2024 average (0.89%), we observe an 80.9% increase in the invalid citation rate, *highlighting 2025 as a dramatic inflection point.* This surge coincides temporally with the widespread adoption of autonomous AI agent workflows capable of generating entire paper sections and references with minimal human oversight. We hypothesize that invalid citations have evolved across three distinct eras: in the pre-LLM period, such errors primarily stemmed from human oversight or citations to restricted-access literature; during the early LLM era (2023–2024), conversational assistants like ChatGPT showed limited impact due to constrained reference-generation capabilities; however, the emergence of agentic workflows in 2025 fundamentally altered this landscape by *enabling fully autonomous content generation, allowing ghost citations to reach research manuscripts at scale.* From a research domain perspective, AI/ML venues exhibit a more pronounced increase in invalid citation counts compared to Security venues, likely reflecting earlier and broader adoption of LLM-based tools within AI/ML research communities.

Repeated Invalid Citations. A particularly concerning phenomenon we observed is "repeated invalid citations": invalid citations (e.g., metadata or title errors) that appear in multiple independent papers. The most notable example is a citation with an erroneous title ("AugMix") that appears in 16 separate papers across AAAI, IJCAI, and NeurIPS (the complete list is in Appendix I). *These viral errors suggest that researchers may be copying citations from other papers that already contain errors,* compounding mistakes across the literature.

> **Archival Analysis**
>
> - *1.07%* of papers contain at least one invalid citation, with a sharp *80.9%* surge in 2025; invalid citations appear across *all venues*.
> - Repeated invalid citations propagate across papers (e.g., the same erroneous title appears in 16 papers), indicating systematic error diffusion.

VII. Analyzing Human Factors: Survey Findings

To understand why ghost citations reach publication despite awareness of the problem, we conducted a user study, distributing surveys and receiving 97 responses.

Statistics Overview. We included paired reverse-worded BibTeX verification items to flag inconsistent responses. After removing 3 conflicting samples, we analyzed 94 valid responses from researchers, with a diverse range of career stages, research areas, and publication experience (see Figure 6). We summarizes our key questions on AI adoption, reporting behavior, peer review efficacy, and support for automated checks in Figure 7, and discuss the main findings and implications below. The complete question wording, numbering and full response breakdown is illustrated in Section J.

A. AI Adoption and Workflow Integration

Our survey reveals widespread adoption of AI tools in academic research workflows. Among respondents who answered the AI-use question (n=86), *87.2%* (75/86) report using AI-powered tools for research purposes. This high adoption rate spans all career stages and research areas, indicating that AI assistance has become normalized in contemporary academic practice.

AI Adoption and Writing Assistance. Among AI users, the majority employ these tools for text polishing: 46.7% use AI for "specific difficult paragraphs," 29.3% use it for "almost every sentence," 22.7% reserve it for "final proofreading" only, and only 1.3% (1/75) report not using AI for polishing. This pervasive integration into the writing process creates multiple opportunities for AI-generated content (including citations) to enter manuscripts.

Metadata Source Practices. When preparing references, most respondents rely on Google Scholar's "Cite" button (72.6%), while 17.9% export directly from publishers (e.g., IEEE/ACM), and 4.8% copy BibTeX entries from other papers. These behaviors increase exposure to propagation risks when upstream metadata contains errors.

B. The Verification Gap

A critical finding emerges when examining verification behavior across both authors and reviewers. While *86.7%* of AI users (65/75) report always verifying AI-generated references, behavioral evidence reveals substantial verification gaps among both groups.

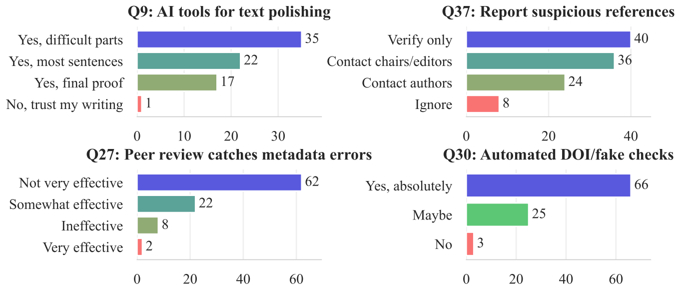


Fig. 7: Key questions on AI adoption, reporting behavior, peer review efficacy, and support for automated checks.

Risky Citation Behaviors. Despite claimed verification, *41.5%* (39/94) copy-paste BibTeX without checking, and *17.3%* (13/75) cite AI-suggested papers without reading them. Among reviewers (n=30), *76.7%* do not thoroughly check references, indicating a trust-by-default norm and a persistent gap between stated verification and actual practice.

Hallucination Experience. Among AI tool users, *41.3%* (31/75) report encountering hallucinated citations "often" or "very often". This high exposure rate, combined with inconsistent verification, creates fertile ground for fabricated citations to propagate into published work. Respondents also view peer review as weak protection: 66.0% rate citation error detection as "not very effective," and 8.5% rate it "ineffective" (74.5% combined). Among reviewers who answered the suspicion question (n=30), 80.0% report never suspecting fake or hallucinated references in submissions. Reporting behavior is also limited (multi-select responses): 44.4% choose no-action options (verify privately or ignore) when encountering suspicious references in published papers. Figure 7 summarizes these key questions spanning AI adoption, reporting behavior, peer review efficacy, and support for automated checks.

C. Perception of Severity and Responsibility

The research community demonstrates awareness of citation integrity issues. *76.6%* of respondents (72/94) consider hallucinated citations a "major problem" or "critical crisis," with 44.7% rating it as a critical crisis and 31.9% as a major problem. This majority recognition of the problem's severity suggests that the persistence of hallucinated citations is not due to ignorance but rather to systemic factors in research workflows.

Responsibility Attribution. When asked who bears primary responsibility for citation accuracy, *91.5%* (86/94) attribute responsibility to authors, while only 3.2% assign it to reviewers, 2.1% to publishers, and 2.1% to AI tool developers. This overwhelming attribution to authors, while arguably correct, may paradoxically reduce pressure on other stakeholders (venues, tool developers) to implement safeguards.

Support for Automated Solutions. Encouragingly, *70.2%* (66/94) strongly support deploying automated DOI/reference checking in submission systems, with an additional *26.6%* responding "maybe." Only 3.2% oppose such measures. This strong support indicates community readiness for technological interventions to address citation integrity.

User Study

- Researchers often take no action on suspicious references (53.3%, 48/90); reviewers do not thoroughly check references (76.7%, 23/30) and rarely suspect fakes (80.0%, 24/30).
- Peer review is viewed as ineffective at catching citation errors (74.5%, 70/94), whereas support for automated checks is overwhelming (70.2%, 66/94).

VIII. Discussion and Mitigation

Through our comprehensive study on citation validity, we show that ghost citations constitute a systemic threat to scholarly trust: fabricated references can enter and persist in the literature because existing verification practices are sparse and largely trust-based. Below, we characterize the lifecycle of this threat and propose targeted interventions.

A. The Lifecycle of Ghost Citations

Our studies collectively map a "pollution pipeline" through which ghost citations propagate from LLM outputs to the permanent scientific record. Understanding this pipeline is essential for designing effective interventions at each stage.

Stage 1: Generation. LLMs do not retrieve citations from verified databases; instead, they "stitch" together plausible combinations of real author names, terminology, and venues, producing ghost citations that are syntactically perfect yet factually non-existent.

Stage 2: Adoption. The high surface plausibility of ghost citations exploits researchers' cognitive offloading, where they trust automated outputs without verification, performing only superficial checks (e.g., confirming the title "looks reasonable") rather than rigorous validation.

Stage 3: Review Failure. Reviewers lack the capacity and time to verify each citation; moreover, academic trust protocols assume authors cite in good faith, so reviewers rarely question whether cited works exist. This combination allows invalid citations to pass through review undetected.

Stage 4: Publication and Propagation. Once published, ghost citations become permanent fixtures. Our discovery of "repeated invalid citations" (appearing in up to 16 papers) reveals that researchers copy citations from existing papers, compounding errors across the literature.

Together, these four stages form a self-reinforcing pipeline that creates *an illusion of evidential support* while contaminating the citation graph with phantom nodes and invalid edges. Left unchecked, this contamination threatens to fundamentally undermine scholarly communication: as fabricated citations accumulate, the research community may be forced to shift from its foundational presumption of trust to one of systematic skepticism, imposing unsustainable verification burdens on every researcher. *We urgently call for coordinated action to* *disrupt this pipeline at multiple points* before the problem becomes intractable.

B. Mitigation and Recommendations

Based on our findings, we propose interventions targeting each stakeholder group. *Addressing ghost citations requires coordinated action* across the entire publication ecosystem; no single stakeholder can solve this problem in isolation. Effective mitigation demands disrupting the pollution pipeline at multiple stages, from generation to publication. Below, we outline actionable recommendations for researchers, venue organizers, AI developers, and the research community.

For Researchers. Verify every AI-generated output before adoption, whether citations, literature summaries, or draft text. Our data shows that even high-performing models fabricate, so we recommend retrieval-grounded tools over purely generative ones and heightened caution when AI output falls outside verifiable domains (e.g., recent citations, unfamiliar subfields). At minimum, check each cited title in a trusted index and treat missing DOIs or inconsistent metadata as red flags; avoid copy-pasting BibTeX entries without verification. Moreover, we urge researchers not to cite any paper—whether suggested by an AI tool or found through traditional search—without having read at least its abstract or a direct summary from the source. Relying solely on AI-generated summaries or metadata risks propagating both fabricated citations and mischaracterized claims.

For Conference and Journal Organizers. Deploy automated citation verification in submission systems (our survey shows 70.2% strong support), require authors to attest that citations have been verified, provide lightweight verification tools to reviewers, and establish clear policies on AI-assisted citation generation. Require structured reference metadata at submission (e.g., BibTeX) and provide reviewers with a compact citation-risk summary to enable targeted checks without increasing burden. For security venues specifically, ghost citations pose unique risks: a fabricated reference in a threat-modeling paper can introduce phantom attack vectors, while an invalid citation in a vulnerability analysis can misattribute credit or obscure the true provenance of a defensive technique. We urge security conference organizers to treat automated citation verification as part of the broader integrity infrastructure, alongside artifact evaluation and reproducibility checks.

For AI Tool Developers. Ground outputs in retrieval from verified sources when factual accuracy matters. Clearly distinguish retrieved from generated content and prompt users to verify before finalizing. Enforce structured outputs with evidence fields (DOI, URL, or database identifier), and surface ”not found” signals instead of fabricating metadata.

For the Research Community. Develop shared verification infrastructure and norms for AI-assisted bibliography construction, invest in detection research, and conduct regular measurement studies to track trends. Build open validation APIs and benchmark datasets, and standardize reporting so policies and tools can evolve with evidence.

C. Limitations

We conducted the first comprehensive, multi-dimensional investigation of ghost citations in the LLM era, developing an automated verification framework and analyzing over 2.2 million citations alongside systematic LLM benchmarking and user surveys. While our work provides critical empirical evidence of this emerging threat, we acknowledge that our study may have several limitations:

Online/CoT Configuration Limits. We enabled online-search/chain-of-thought via third-party API flags. These may not activate vendors' full native toolchains, so the online-vs-offline comparison reflects interface-level settings rather than definitive vendor performance.

Detection and Verification Limitations. Our pipeline verifies only title similarity, reflecting our goal of detecting *ghost citations* (references that cannot be traced or do not exist) rather than *citation errors*. This conservative approach may undercount hallucinations that closely resemble real papers, while some flagged citations may be legitimate but poorly indexed works; our manual verification addresses the latter, but false negatives remain a limitation. *The detected invalid citations are likely to be a conservative lower bound* on the true hallucination rate. To mitigate human bias in manual verification, each flagged citation was double-checked by independent reviewers to ensure accuracy. However is still possible that some internal or poorly indexed papers were misclassified as invalid citations.

External Validity of LLM Benchmark. Our LLM benchmark employs a controlled, standardized prompting protocol in which models generate fixed-size blocks of citations per domain. While this design ensures internal validity—enabling fair cross-model and cross-domain comparisons under identical conditions—it may not fully capture the diversity of real-world researcher workflows, where citations are generated in the context of specific drafts, arguments, or literature reviews rather than via bulk requests. We therefore caution against interpreting our hallucination rates as direct estimates of real-world prevalence; rather, they represent a rigorous lower bound under standardized conditions. Future work could strengthen external validity by observing or replicating realistic citation-generation scenarios, such as assisting with actual paper drafts.

Survey Response Biases. Our survey relies on self-reported behavior, which is susceptible to social-desirability bias: respondents may over-report diligent practices (e.g., verifying citations) and under-report risky behaviors (e.g., copy-pasting without checking) [52]. The substantial gap we observe between stated diligence (86.7% claim to "always verify") and admitted risky behavior (41.5% copy-paste BibTeX without checking) is itself suggestive of this bias. While paired reverse-worded items (Q38/Q39) allowed us to flag and exclude three inconsistent responses, they cannot eliminate systematic social-desirability effects. Furthermore, demographic questions were placed at the beginning of the survey, which prior work suggests may prime respondents toward socially-desirable responses on subsequent items [52]. We did not conduct cognitive interviews or expert questionnaire reviews during pre-testing, both of which are recommended best practices for identifying leading or ambiguous items [52]. We therefore interpret our survey findings as indicative of perceived norms and self-reported tendencies rather than definitive measurements of actual behavior.

IX. Conclusion

We presented the first comprehensive investigation of citation validity in the age of LLMs, mapping how ghost citations flow from generation through adoption, review failure, and propagation. We develop CITEVERIFIER as an open-source tool that establishes a measurement baseline and clarifies the systemic nature of the threat. We hope our work motivates the community to treat citation integrity as a critical priority and call for immediate efforts to safeguard the scholarly record before ghost citations become an intractable problem.

Ethical Considerations

In this section, we discuss the ethical considerations of our study and how they were addressed in our implementation of CITEVERIFIER and the associated measurements.

Responsible Use of External Services. All LLMs and external services were accessed in compliance with their respective terms of service and usage guidelines. We applied rate limiting and conservative request schedules to avoid overloading provider infrastructure, and we did not use adversarial or harmful prompts, following the Menlo report [18].

Data Access and Stewardship. Paper collection relied on publicly accessible proceedings and institutionally authorized access by our research team. For open-access venues, we used DBLP to obtain DOI links and downloaded papers via automated scripts; for venues that provide full proceedings, we downloaded the proceedings and split PDFs by the table of contents; for access-restricted venues, we used automated browser control to mimic manual downloads under institutional access. We enforced a 5-second delay between downloads to avoid server burden, and any access-restricted papers were retrieved only through the researchers' institutional access rights. All papers are used solely for academic research, and no portion of the corpus has been redistributed or disseminated to the outside.

Human Subjects and Privacy. The survey received IRB approval from our institution prior to data collection. Participants were recruited through two transparent channels: we published the survey link on social media platforms, and we randomly sampled 300 researchers from program committees and author lists of top-tier venues for direct email outreach through their institutional or publicly available email addresses. The survey provided full transparency to participants, including detailed explanation of the research purpose, methodology, contact information for the research team, and an explicit opt-out mechanism allowing withdrawal at any point. The questionnaire did not solicit personally identifiable information; all responses were stored securely in de-identified form and have not been shared outside the research.

Anonymization of Papers with Invalid Citations. Following the precedent of meta-science studies that critically examine published work [23], we decided not to disclose the identities of specific papers containing invalid citations in the submitted manuscript. Our goal is not to blame individual authors, institutions, or papers, but to inform and spark constructive discussion around a widespread phenomenon: the presence of ghost citations (untraceable or fabricated references) in the scientific record, that affects the entire research community. Invalid citations may arise for various reasons (e.g., researcher oversight, uncritical adoption of LLM-generated content, or gaps in verification workflows) and do not necessarily constitute evidence of intent. We acknowledge that causality and attribution are complex; our focus is on systemic patterns rather than individual cases. We welcome feedback from reviewers on how best to balance transparency with proportionality in addressing this emerging challenge.

Acknowledgment

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TABLE V: Large language models evaluated in citation hallucination benchmark (13 models across major vendors).

Model	**Vendor**	**Release Date**
GPT-5	OpenAI	2025-08-07
Claude-Sonnet-4	Anthropic	2025-05-22
Gemini-2.5-Pro	Google DeepMind	2025-06-17
Grok-4-Fast	XAI	2025-09-19
Llama-4-Maverick	Meta	2025-04-05
Phi-4	Microsoft	2024-12-12
GLM-4.5	Zhipu AI	2025-07-28
Hunyuan-A13B-Instruct	Tencent	2025-06-27
UI-TARS-1.5-7B	ByteDance Seed	2025-04-17
Qwen3-Max	Alibaba Qwen	2025-09-23
DeepSeek-Chat-V3.1	Deepseek	2025-08-21
ERNIE-4.5-21B-A3B	Baidu	2025-06-30
Kimi-K2-0905	Moonshot	2025-09-05

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APPENDIX

In the appendix, we provide additional details on our study that are not included in the main text due to space constraints.

A. Models Evaluated

Table V lists the 13 LLMs used in our benchmark and their access interfaces.

B. Research Domain Codes

Table X provides the mapping between domain abbreviation codes and their full names, as used in the LLM citation generation experiment (Section V). These 40 domains cover the major areas of computer science as categorized by arXiv.

C. LLM Citation Generation Prompt

We prompt the model to generate citations using the following prompt:

> You are a senior academic research assistant. Please complete the task according to the following requirements. [Requirements] 1. Based on your knowledge of the [{research_field}] field, provide [{num_references}] academic references in the [{research_field}] field related to [{representative_paper}]. 2. Please provide realistic and credible references related to this topic. 3. Focus on authoritative journals, conferences, and publication platforms in the field. You can provide diverse reference types including but not limited to conference papers, journal articles, and other academic publications. 4. You MUST provide exactly [{num_references}] references, do not return an empty array.
> [Output Requirements] - You MUST output ONLY valid JSON format, no explanations, no additional text, no apologies - Do NOT include any text before or after the JSON array - You MUST provide exactly [{num_references}] references - Your entire response must be parseable as JSON
> [Output Format] Output ONLY the following JSON format: [{ "author": ["Author1", "Author2", "Author3"], "title": "Full Article Title", "venue": "Name of the journal, conference, or publication platform", "year": Publication year, "url": "Article link (if any)", "doi": "DOI number (if any)", "reference_type": "Reference type" }]
> [Field Requirements] - author: Array of author name strings, including all author names (required) - title: Full article title (required) - venue: Name of the published journal, conference, or platform (required) - year: Publication year, in numeric format (required) - url: Article access link, fill in null if none - doi: DOI number, fill in null if none - reference_type: Reference literature type, choose from: article (conference/journal papers), series (book series), thesis (degree theses), monograph (books), unknown (when type cannot be determined)
> REMEMBER: Output ONLY JSON with exactly [{num_references}] references, no other text whatsoever.

D. LLM Citation Validity Judgment Prompt

The following prompt was used in our auxiliary experiment where each model judged whether a bibliographic entry corresponds to a real publication.

> You are a rigorous academic fact-checking assistant. You will receive one bibliographic entry with fields: Cite Title, Authors, Year, Venue. Decide whether it corresponds to a real academic publication where the combination of title, authors, year, and venue is accurate.
> [Task] - Judge a single entry. - Output ONLY a JSON object with exactly two fields: "result" and "reason". - "result" should be true or false (lowercase, JSON boolean). If you judge the entry corresponds to a real publication, output true; otherwise output false. - "reason" should be a short reason (1-3 sentences). - No extra keys, no markdown, no surrounding text.
> Entry: Cite Title: title Authors: authors Year: year Venue: venue

TABLE VI: Citation hallucination rates: offline vs. online+thinking. Δ shows the difference between two modes.

Model	Offline (%)	Online (%)	Δ (%)
Seed	75.78	94.50	+18.72
GPT-5	42.54	58.97	+16.42
Llama 4	44.22	47.50	+3.27
Claude 4	21.43	22.25	+0.83
GLM-4.5	21.20	21.29	+0.10
Phi-4	87.47	87.48	+0.02
Gemini	59.67	59.26	-0.42
Grok 4	80.26	79.70	-0.57
Qwen-3	23.83	23.21	-0.62
Hunyuan	95.27	94.59	-0.68
Kimi	42.28	41.45	-0.84
DeepSeek	14.68	13.78	-0.89
ERNIE	75.77	68.12	-7.65
Average	52.65	54.78	**+2.13**

E. Factors Affecting Hallucination Rates

In this section, we provide additional analysis on factors that may influence citation hallucination rates in LLMs, including the impact of online search and chain-of-thought prompting, batch size effects, and domain-level sensitivity.

a) Online Search and Chain-of-Thought Impact: Table VI reports citation hallucination rates with online search and chain-of-thought prompting versus baseline settings, along with per-model differences. These settings are controlled through a third-party API aggregation interface, so results reflect the API-level toggles we could access rather than each vendor's full native retrieval or reasoning stack.

b) Batch Size Impact on Hallucination Rates: Table IX presents the detailed breakdown of citation hallucination rates by batch size (10, 20, and 30 citations per prompt) for all 13 models. The results show no consistent correlation between batch size and hallucination rates, suggesting that hallucination reflects fundamental limitations in the models' bibliographic knowledge rather than output quantity constraints.

F. Domain-Level Hallucination Rates

Table VII presents citation hallucination rates by research domain (40 CS subfields), aggregated across all 13 models and ranked from highest to lowest.

G. Citation Stability and Cross-Model Overlap

We provide additional analysis of citation stability across repeated runs and cross-model overlap patterns. Table VIII reports stability statistics for valid versus hallucinated citations, while Figure 8 visualizes pairwise overlap of generated citation sets using the Extended Jaccard similarity (percentage). Diagonal cells are 100% by definition, and off-diagonal values indicate how similar two models' generated citations are for the same prompts. Higher overlap suggests more similar citation preferences across models, while lower overlap indicates greater divergence in the generated reference sets under identical prompts. We observe that some model

TABLE VII: Citation hallucination rates (%) by research domain (40 CS subfields), ranked from highest to lowest.

Domain	Halluc.	95% CI	Domain	Halluc.	95% CI
DL (Digital Libraries)	80.19	±3.46	FL (Formal Languages)	49.60	±4.32
OH (Other Hardware)	75.38	±3.73	DM (Discrete Math)	49.33	±4.30
NI (Network Info.)	73.42	±3.75	PL (Prog. Languages)	48.40	±4.28
ET (Emerg. Tech.)	69.38	±3.97	OS (Operating Sys.)	47.96	±4.32
GT (Game Theory)	64.04	±4.10	SC (Sound)	47.23	±4.28
SY (Symbolic Comp.)	63.90	±4.12	NE (Networking)	45.58	±4.26
CY (Computers & Soc.)	62.24	±4.22	SD (Software Dev.)	45.51	±4.34
AR (Architecture)	60.59	±4.21	NA (Num. Analysis)	44.01	±4.27
CE (Comp. Eng.)	60.58	±4.22	DC (Distributed Comp.)	43.96	±4.29
DB (Databases)	60.31	±4.23	IT (Info. Theory)	43.18	±4.23
SE (Software Eng.)	60.26	±4.19	LG (Learning)	41.92	±4.30
CR (Cryptography)	59.96	±4.20	LO (Logic in CS)	41.55	±4.28
HC (Human-Comp. Int.)	59.51	±4.27	MS (Math. Software)	35.03	±4.11
MM (Multimedia)	59.20	±4.23	CV (Comp. Vision)	33.64	±4.08
AI (Artificial Intel.)	57.32	±4.23	GR (Graphics)	32.50	±4.07
PF (Performance)	54.63	±4.31	CC (Comp. Complexity)	32.23	±4.02
MA (Multiagent Sys.)	53.97	±4.30	GL (General Lit.)	31.50	±4.03
RO (Robotics)	52.21	±4.37	DS (Data Structures)	29.76	±3.93
IR (Info. Retrieval)	51.05	±4.29	SI (Social/Info. Net.)	29.31	±3.92
CG (Comp. Geometry)	49.66	±4.31	CL (Comp. Linguistics)	28.80	±4.00

Halluc.: Hallucination rate, the percentage of citations verified as invalid.

TABLE VIII: Citation stability scores across repeated runs. Sorted by valid citation stability (higher means more consistent).

Model	Valid Stability		Halluc. Stability	
	Mean	Std	Mean	Std
DeepSeek	0.577	0.094	0.226	0.216
Qwen-3	0.574	0.090	0.175	0.199
Claude 4	0.571	0.187	0.181	0.187
Gemini	0.552	0.156	0.069	0.083
GLM-4.5	0.528	0.115	0.190	0.203
GPT-5	0.512	0.117	0.040	0.036
Llama 4	0.469	0.150	0.097	0.070
Grok 4	0.380	0.194	0.035	0.027
Hunyuan	0.348	0.255	0.024	0.040
Kimi	0.341	0.090	0.041	0.080
ERNIE	0.323	0.164	0.034	0.040
Seed	0.246	0.191	0.047	0.091
Phi-4	0.191	0.146	0.007	0.012

Halluc. Stability: consistency of hallucinated citations across runs;

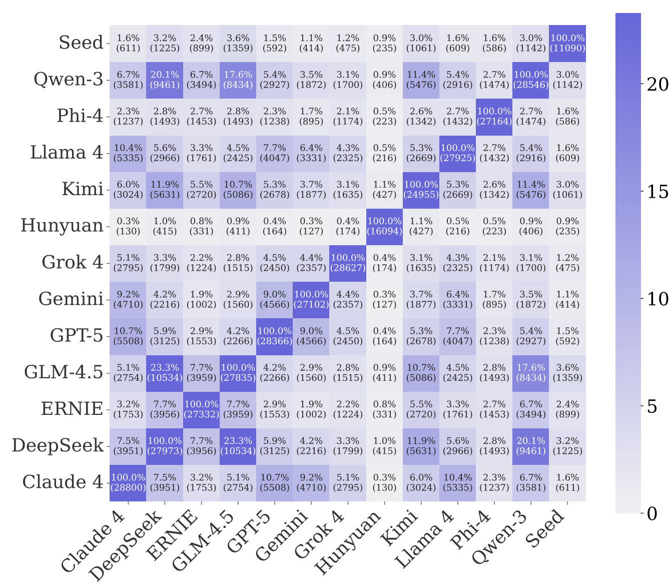

Fig. 8: Cross-model citation overlap heatmap. Color represents the Extended Jaccard similarity percentage.

pairs (e.g. GLM4.5, DeepSeek, Qwen-3) exhibit comparatively higher overlap, indicating closer citation preferences.

Table XI presents the most frequently generated valid citations for each of the 40 research domains in our LLM benchmark experiment. These papers represent the references that LLMs most consistently recall, suggesting they are strongly encoded in the models' parametric memory due to high frequency in training data. Seminal works such as "NeRF" (Graphics), "RAG" (Computation & Language), and "U-Net" (Computer Vision) appear at the top, reflecting the influence of foundational papers in each field.

H. Paper Collection Statistics

The complete breakdown of papers collected by conference and year (2020–2025) is shown in Table III in the main text.

I. Repeated Invalid Citations in Published Papers

Table XII presents invalid citations (e.g., metadata or title errors) that appear in multiple independent published papers across our corpus of 56,381 papers. These "repeated invalid citations" demonstrate how citation mistakes propagate when

TABLE IX: Citation hallucination rates (%) by batch size (10, 20, 30 references). Sorted by range (most stable first).

Model	10 refs	20 refs	30 refs	Range
Gemini	59.74	59.06	59.22	0.68
GLM-4.5	20.91	21.84	21.39	0.94
Kimi	42.22	41.77	40.78	1.45
Phi-4	87.83	87.46	86.31	1.52
Grok 4	79.84	81.15	78.64	2.51
Hunyuan	94.61	94.55	97.11	2.56
Qwen-3	24.39	22.95	21.72	2.68
Claude 4	23.00	20.11	20.95	2.89
Llama 4	45.77	44.76	47.70	2.94
DeepSeek	15.31	13.26	12.36	2.96
Seed	79.01	77.30	84.27	6.97
ERNIE	75.49	65.56	70.71	9.94
GPT-5	44.07	59.95	57.24	15.87
Average	53.25	53.05	53.72	–

researchers copy references from papers that already contain errors. The most frequently occurring invalid citation (a malformed title, e.g., "Augmix" instead of "AugMix") appears in 16 separate papers across IJCAI, NeurIPS, and AAAI. Notably, "DeepIP" appears across both AI and Security venues (USENIX, CCS, NeurIPS), indicating cross-domain propagation of citation errors.

J. Survey Response Details (Full)

Tables XIV to XVII provide the complete numerical breakdown of all survey responses (N=94 valid responses, 97 total), including every response option for adoption, verification, severity, responsibility, support for automated tools, risky behaviors, reviewer reactions, text polishing, primary metadata source, and claim verification frequency. The question wording and numbering map is shown in Table XIII.

TABLE X: Research domain codes used in the LLM citation generation experiment.

Code	Research Domain	Code	Research Domain
AI	Artificial Intelligence	IT	Information Theory
AR	Hardware Architecture	LG	Machine Learning
CC	Computational Complexity	LO	Logic in Computer Science
CE	Computational Engineering, Finance, and Science	MA	Multiagent Systems
CG	Computational Geometry	MM	Multimedia
CL	Computation and Language	MS	Mathematical Software
CR	Cryptography and Security	NA	Numerical Analysis
CV	Computer Vision and Pattern Recognition	NE	Neural and Evolutionary Computing
CY	Computers and Society	NI	Networking and Internet Architecture
DB	Databases	OH	Other Computer Science
DC	Distributed, Parallel, and Cluster Computing	OS	Operating Systems
DL	Digital Libraries	PF	Performance
DM	Discrete Mathematics	PL	Programming Languages
DS	Data Structures and Algorithms	RO	Robotics
ET	Emerging Technologies	SC	Symbolic Computation
FL	Formal Languages and Automata Theory	SD	Sound
GL	General Literature	SE	Software Engineering
GR	Graphics	SI	Social and Information Networks
GT	Computer Science and Game Theory	SY	Systems and Control
HC	Human-Computer Interaction		
IR	Information Retrieval		

TABLE XI: Most frequently generated valid citations by research domain across all 13 LLMs.

Topic	Most Cited Paper Title	Count	%
GR	NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis	352	6.46
CL	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	348	6.24
NI	Green Traffic Engineering for Satellite Networks Using Segment Routing Flexible Algorithm	138	5.97
ET	FCT O-RAN: Design and Deployment of a Multi-Vendor End-to-End Private 5G Testbed	148	5.69
AR	Eyeriss: An Energy-Efficient Reconfigurable Accelerator for Deep Convolutional Neural Networks	184	5.55
DL	Digital Libraries	96	5.55
IT	A Mathematical Theory of Communication	260	5.46
SI	Semi-Supervised Classification with Graph Convolutional Networks	287	4.88
CV	U-Net: Convolutional Networks for Biomedical Image Segmentation	262	4.83
GT	Algorithmic Game Theory	146	4.62
DC	MapReduce: Simplified Data Processing on Large Clusters	202	4.47
AI	Language Models are Few-Shot Learners	154	4.28
SD	Musical genre classification of audio signals	185	4.27
CG	Computational Geometry: Algorithms and Applications	178	4.19
SE	Automated Generation of Issue-Reproducing Tests by Combining LLMs and Search-Based Testing	141	4.14
FL	Introduction to Automata Theory, Languages, and Computation	171	4.10
GL	Denoising Diffusion Probabilistic Models	234	4.08
CC	Computational Complexity: A Modern Approach	230	4.07
HC	AttenTrack: Mobile User Attention Awareness Based on Context and External Distractions	130	3.93
MA	Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations	152	3.89
MM	LLM-Guided Semantic Relational Reasoning for Multimodal Intent Recognition	134	3.83
PL	Dato: A Task-Based Programming Model for Dataflow Accelerators	168	3.80
CY	Algorithms of Oppression: How Search Engines Reinforce Racism	114	3.56
OS	Scheduling Algorithms for Multiprogramming in a Hard-Real-Time Environment	153	3.51
DS	Introduction to Algorithms	202	3.42
OH	Cutting the Electric Bill for Internet-Scale Systems	71	3.34
MS	Robust Regression and Outlier Detection	180	3.25
PF	Optimal Parallel Scheduling under Concave Speedup Functions	122	3.25
CE	A continuum multi-species biofilm model with a novel interaction scheme	103	3.15
LO	Language-Based Information-Flow Security	145	3.03
LG	Causality: Models, Reasoning, and Inference	140	2.92
SY	Three-Phase PLLs: A Review of Recent Advances	81	2.66
IR	BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer	110	2.65
CR	A Method for Obtaining Digital Signatures and Public-Key Cryptosystems	91	2.61
NA	Numerical Linear Algebra	121	2.56
DM	Polyphase codes with good periodic correlation properties	107	2.50
SC	Planning Algorithms	110	2.45
RO	Human-Robot Interaction: A Survey	96	2.44
NE	Handbook of Evolutionary Computation	114	2.42
DB	MapReduce: Simplified Data Processing on Large Clusters	71	2.08

TABLE XII: Repeated invalid citations: invalid citations appearing in multiple published papers across top-tier venues.

Invalid Citation Title	Count	Venues
Augmix: A Simple Method To Improve Robustness And Uncertainty Under Data Shift	16	AAAI, IJCAI, NeurIPS
Reduction Of A Game With Complete Memory To A Matrix Game	7	AAAI, NeurIPS
Towards Private Synthetic Text Generation	5	ICML
Turing: Composable Inference For Probabilistic Programming	3	ICML
DeepIP: Deep Neural Network Intellectual Property Protection With Passports	3	USENIX, CCS, NeurIPS
A Multi-Illumination Dataset Of Indoor Object Appearance	3	ICML, NeurIPS
On The Systematic Fitting Of Frequency Curves	3	NeurIPS
Keeping Neural Networks Simple By Minimising The Description Length Of The Weights	3	ICML, NeurIPS
Algorithm Cnneim-A And Its Mean Complexity	3	ICML, NeurIPS
Truncated Horizon Policy Search: Deep Combination Of Reinforcement And Imitation	2	ICML, NeurIPS

TABLE XIII: Survey question numbering map.

Q#	Question
Q1	What is your current academic position?
Q2	Which research area best describes your work?
Q3	How many peer-reviewed papers have you published in your career?
Q4	Have you served as a reviewer for top-tier conferences (e.g., USENIX Sec, CCS, S&P, CVPR, NeurIPS) in the last 3 years?
Q5	Which stage of paper writing do you find most time-consuming? (Select up to 2)
Q6	How do you typically search for related work?
Q7	Do you use AI-powered tools for research?
Q8	Which types of AI-powered tools do you use for research? (Select all that apply)
Q9	Do you use AI tools specifically for text polishing, grammar checking, or rephrasing?
Q10	Specifically regarding citations, how do you use AI tools? (Select all that apply)
Q11	When asking a general LLM (e.g., ChatGPT) to find references, how often do you encounter hallucinations (non-existent papers)?
Q12	If an AI tool provides a perfect-looking reference (Title, Author, Year, Venue all look correct), do you still verify it externally?
Q13	Have you ever cited a paper suggested by AI without reading the full text of that paper?
Q14	To what extent do you agree: AI tools have made it easier to generate related work sections, but harder to ensure citation accuracy.
Q15	What is your strategy when an AI tool generates a citation with a broken link or DOI?
Q16	When adding a citation to your paper, what is your primary source of metadata (title, year, venue)?
Q17	How often do you verify that a cited paper actually contains the claim you are attributing to it?
Q18	If you cannot find the full text of a paper (e.g., paywalled or offline), do you still cite it based on its abstract or title?
Q19	To what extent do you agree: There is pressure to include a high number of references to make the paper look more scholarly.
Q20	When reviewing a paper, do you explicitly check the Reference section?
Q21	What are you looking for when checking references? (Select all that apply)
Q22	Have you ever clicked on a DOI link in a submission's bibliography to verify the paper exists?
Q23	Have you ever suspected a submission contained fake or hallucinated references?
Q24	If you see a citation to a preprint or ArXiv paper, do you verify if it has been published in a peer-reviewed venue?
Q25	How do you handle citations to non-English papers in a submission?
Q26	To what extent do you agree: The accuracy of the bibliography is as important as the accuracy of the experimental results.
Q27	Do you believe the current peer review process is effective at catching metadata errors in references?
Q28	How serious of an issue do you consider hallucinated citations (non-existent papers) in the era of AI?
Q29	Who is primarily responsible for verifying the authenticity of cited references?
Q30	Should conferences deploy automated tools to check for broken DOIs or fake references upon submission?
Q31	To what extent do you agree: It is acceptable to cite a paper based on AI summarization without reading the original text.
Q32	Have you ever seen a reference list where the author names were clearly scrambled or incorrect (e.g., order reversed)?
Q33	Have you ever seen a reference where the title existed but the venue/year was completely wrong?
Q34	Do you think it is acceptable to cite a technical report if a peer-reviewed version exists?
Q35	What is your reaction if you find one incorrect citation (e.g., wrong year) in a paper you are reviewing?
Q36	What is your reaction if you find multiple (e.g., ¿5) incorrect or fake citations?
Q37	If you encounter a suspicious or clearly fake reference in a published paper, do you report it?
Q38*	Have you ever copy-pasted a BibTeX entry from the internet without checking its content?
Q39*	To what extent do you agree: I meticulously verify every single field (volume, issue, page numbers) of every BibTeX entry I import, ensuring 100% accuracy before submission.
Q40	Do you use tools like Semantic Scholar or ResearchRabbit to build your bibliography?

Reverse-worded consistency check: Q38 "risky behavior" vs. Q39 "diligent behavior." Three respondents answered inconsistently ("often copy-paste" AND "strongly verify") and were excluded.

TABLE XIV: Full survey response details—Part A-1: Q1–Q10 (N=94 valid responses).

Item	N	%	Item	N	%
Q1: What is your current academic position?			*Q2: Which research area best describes your work?*		
PhD Student	32	34.0	Other	24	21.1
Master's Student	19	20.2	AI	20	17.5
Faculty (Professor/Lecturer)	17	18.1	Machine Learning	20	17.5
Undergraduate student	13	13.8	Network Security	18	15.8
Postdoc	6	6.4	AI Security	13	11.4
Researcher	5	5.3	System Security	10	8.8
Other	2	2.1	Cryptography	5	4.4
			Software Engineering	4	3.5
Q3: How many peer-reviewed papers have you published in your career?			*Q4: Have you served as a reviewer for top-tier conferences (e.g., USENIX Sec, CCS, S&P, CVPR, NeurIPS) in the last 3 years?*		
1–5	46	48.9	No	62	66.0
6–10	16	17.0	Yes	32	34.0
0	14	14.9			
20+	12	12.8			
11–20	6	6.4			
Q5: Which stage of paper writing do you find most time-consuming? (Select up to 2)			*Q6: How do you typically search for related work?*		
Conceptualization	62	35.4	Keyword search on Google Scholar/DBLP	68	33.5
Writing the Introduction	36	20.6	Following citations from other papers	62	30.5
Literature Search	24	13.7	Browsing conference proceedings	40	19.7
Review	24	13.7	Using AI-powered tools (e.g., ChatGPT, Connected Papers)	33	16.3
Technical Description	24	13.7			
Formatting References	5	2.9			
Q7: Do you use AI-powered tools for research?			*Q8: Which types of AI-powered tools do you use for research? (Select all that apply)*		
Yes	75	87.2	General-purpose LLMs (ChatGPT, Claude, Gemini)	75	58.6
No	11	12.8	Coding Assistants (Copilot, Cursor)	41	32.0
			Academic search (Elicit, Semantic Scholar)	6	4.7
			AI reading tools (ChatPDF, Humata)	6	4.7
Q9: Do you use AI tools specifically for text polishing, grammar checking, or rephrasing?			*Q10: Specifically regarding citations, how do you use AI tools? (Select all that apply)*		
Yes, for specific difficult paragraphs	35	46.7	None (do not use AI for citation tasks)	35	29.9
Yes, for almost every sentence	22	29.3	Formatting (convert to BibTeX, etc.)	23	19.7
Yes, only for final proofreading	17	22.7	Discovery (recommend papers on topic)	23	19.7
No, trust my own writing more	1	1.3	Summarization (decide whether to cite)	15	12.8
			Gap-filling (need a citation)	15	12.8
			Verification (ask if paper exists)	6	5.1

TABLE XV: Full survey response details—Part A-2: Q11–Q20 (N=94 valid responses).

Item	**N**	**%**	**Item**	**N**	**%**
Q11: When asking a general LLM (e.g., ChatGPT) to find references, how often do you encounter hallucinations (non-existent papers)?			*Q12: If an AI tool provides a perfect-looking reference (Title, Author, Year, Venue all look correct), do you still verify it externally?*		
Often (20–50%)	24	32.0	Always (e.g., Scholar/DBLP)	58	77.3
Don’t use LLMs for finding refs	25	33.3	Only if reading full text	8	10.7
Occasionally (¡20%)	16	21.3	Yes, verify 100% via Scholar/DBLP	7	9.3
Very Often (¿50%)	7	9.3	Only if suspicious	1	1.3
Never	3	4.0	Only if need to read the full text	1	1.3
Q13: Have you ever cited a paper suggested by AI without reading the full text of that paper?			*Q14: To what extent do you agree: AI tools have made it easier to generate related work sections, but harder to ensure citation accuracy.*		
No, never	62	82.7	Somewhat agree	28	37.3
Yes, once or twice	8	10.7	Strongly agree	24	32.0
Yes, often	5	6.7	Neutral	11	14.7
			Somewhat disagree	9	12.0
			Strongly disagree	3	4.0
Q15: What is your strategy when an AI tool generates a citation with a broken link or DOI?			*Q16: When adding a citation to your paper, what is your primary source of metadata (title, year, venue)?*		
Assume exists, find manually	45	60.0	Google Scholar “Cite” button	61	72.6
Assume hallucination, discard	24	32.0	Direct export from publisher	15	17.9
Ask AI for different link	4	5.3	Direct export from publisher (IEEE/ACM)	4	4.8
Keep citation, remove DOI	2	2.7			
Q17: How often do you verify that a cited paper actually contains the claim you are attributing to it?			*Q18: If you cannot find the full text of a paper (e.g., paywalled or offline), do you still cite it based on its abstract or title?*		
Every single time	48	57.1	No, never	46	54.8
Most of the time	20	23.8	Yes, if necessary	35	41.7
Only for critical claims	14	16.7	Yes, often	3	3.6
Rarely	2	2.4			
Q19: To what extent do you agree: There is pressure to include a high number of references to make the paper look more scholarly.			*Q20: When reviewing a paper, do you explicitly check the Reference section?*		
Somewhat agree	31	36.9	Yes, skimming	18	60.0
Neutral	21	25.0	Yes, carefully	7	23.3
Somewhat disagree	17	20.2	No	5	16.7
Strongly disagree	10	11.9			
Strongly agree	5	6.0			

TABLE XVI: Full survey response details—Part B-1: Q21–Q30 (N=94 valid responses).

Item	N	%	Item	N	%
Q21: What are you looking for when checking references? (Select all that apply)			*Q22: Have you ever clicked on a DOI link in a submission's bibliography to verify the paper exists?*		
Missing key related work	28	52.8	Occasionally	13	43.3
Correctness of metadata	10	18.9	Rarely	9	30.0
Existence of papers	10	18.9	Never	6	20.0
Self-citation abuse	5	9.4	Frequently	2	6.7
Q23: Have you ever suspected a submission contained fake or hallucinated references?			*Q24: If you see a citation to a preprint or ArXiv paper, do you verify if it has been published in a peer-reviewed venue?*		
No	24	80.0	Sometimes	15	50.0
Yes	6	20.0	No	9	30.0
			Always	6	20.0
Q25: How do you handle citations to non-English papers in a submission?			*Q26: To what extent do you agree: The accuracy of the bibliography is as important as the accuracy of the experimental results.*		
I check them using translation tools	10	33.3	Strongly agree	52	55.3
I ignore them	10	33.3	Somewhat agree	29	30.9
I assume they are valid	9	30.0	Neutral	10	10.6
I ask the authors to clarify	1	3.3	Somewhat disagree	3	3.2
Q27: Do you believe the current peer review process is effective at catching metadata errors in references?			*Q28: How serious of an issue do you consider hallucinated citations (non-existent papers) in the era of AI?*		
Not very effective	62	66.0	Critical crisis	42	44.7
Somewhat effective	22	23.4	Major problem	30	31.9
Ineffective	8	8.5	Minor nuisance	22	23.4
Very effective	2	2.1			
Q29: Who is primarily responsible for verifying the authenticity of cited references?			*Q30: Should conferences deploy automated tools to check for broken DOIs or fake references upon submission?*		
Authors	86	91.5	Yes, absolutely	66	70.2
Reviewers	3	3.2	Maybe	25	26.6
Publishers (Editorial)	1	1.1	No	3	3.2
AI tool developers	2	2.1			

TABLE XVII: Full survey response details—Part B-2: Q31–Q40 (N=94 valid responses).

Item	**N**	**%**	**Item**	**N**	**%**
Q31: To what extent do you agree: It is acceptable to cite a paper based on AI summarization without reading the original text.			*Q32: Have you ever seen a reference list where the author names were clearly scrambled or incorrect (e.g., order reversed)?*		
Strongly disagree	33	35.1	No	59	62.8
Somewhat disagree	31	33.0	Yes	35	37.2
Neutral	17	18.1			
Somewhat agree	12	12.8			
Strongly agree	1	1.1			
Q33: Have you ever seen a reference where the title existed but the venue/year was completely wrong?			*Q34: Do you think it is acceptable to cite a technical report if a peer-reviewed version exists?*		
No	50	53.2	Yes	72	76.6
Yes	44	46.8	No	22	23.4
Q35: What is your reaction if you find one incorrect citation (e.g., wrong year) in a paper you are reviewing?			*Q36: What is your reaction if you find multiple (e.g., ¿5) incorrect or fake citations?*		
Mention in minor comments	61	64.9	Reject immediately (ethical concern)	56	59.6
Ask for major revision	15	16.0	Ask for explanation	38	40.4
Reject the paper	4	4.3			
Q37: If you encounter a suspicious or clearly fake reference in a published paper, do you report it?			*Q38: Have you ever copy-pasted a BibTeX entry from the internet without checking its content?**		
Verify privately, take no action	40	37.0	No, never	55	58.5
Contact PC Chairs / Journal Editors	36	33.3	Yes, rarely	26	27.7
Contact authors directly	24	22.2	Yes, often	13	13.8
Ignore	8	7.4			
*Q39: To what extent do you agree: I meticulously verify every single field (volume, issue, page numbers) of every BibTeX entry I import, ensuring 100% accuracy before submission.**			*Q40: Do you use tools like Semantic Scholar or ResearchRabbit to build your bibliography?*		
Somewhat agree	27	28.7	No	78	83.0
Strongly agree	26	27.7	Yes	16	17.0
Neutral	26	27.7			
Somewhat disagree	10	10.6			
Strongly disagree	5	5.3			

Q38 and Q39 are semantically opposite questions. Q38 measures risky behavior ("Have you ever copy-pasted a BibTeX entry from the internet without checking its content?"), while Q39 measures diligent behavior ("To what extent do you agree: I meticulously verify every single field (volume, issue, page numbers) of every BibTeX entry I import, ensuring 100% accuracy before submission.").